From Atomic RF Detection to Full-Stokes Microwave Polarimetry: A Reconfigurable L-Band Metasurface for Rydberg Sensing

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A polarisation-aware radio-frequency sensor must distinguish electromagnetic fields having equal intensity but different transverse amplitude and phase. This thesis develops a reconfigurable L-band metasurface that converts incident polarisation into six sequential, linearly independent receiving responses. The work connects an existing optically interrogated rubidium receiver, measured linear-polariser discrimination and a two-copper-face split-loop circuit designed for a single printed substrate. An experimental RF campaign yielded 9.7 dB preferred-to-rejected discrimination near 2 GHz. The electromagnetic device was assessed using Ansys HFSS complex Jones transmission matrices and a physically specified quadratic receiving operator. A 36-candidate, eleven-parameter geometry search increased the longest passing sampled discovery span from 175 to 250 MHz; the latter is a screening-grid result rather than a continuously validated bandwidth. The selected bandwidth-optimised design gives a full-rank six-state Stokes matrix and a normalised condition number of 9.53 at 1.925 GHz in an independently adapted periodic model. Algebraic reconstruction from those HFSS matrices recovers 1,231 prescribed states on the Poincaré sphere to floating-point precision. Separate finite 3×3-panel simulations recover a full-rank six-state receiving operator at 1.850 GHz and reveal the importance of receiver distance and mesh resolution. We derive the Jones-to-Stokes operator, an observability criterion, a circular-response identity and aperture-scaling diagnostics, and retain traceable solver and figure data. The results establish a reproducible electromagnetic front-end concept for future atom-based microwave polarimetry; optical full-Stokes readout and physical diode switching remain future integration steps.

Keywords: Rydberg-atom sensing; microwave polarimetry; L-band; switchable metasurface; Stokes vector; Poincaré sphere; HFSS

NOMENCLATURE AND ACRONYMS

$\boldsymbol{S}$	Stokes four-vector (S_0, S_1, S_2, S_3)
J_k	complex Jones transmission matrix for mask k
A	Stokes instrument matrix assembled across masks
κ_2	spectral condition number
Ω_{RF}	microwave Rabi frequency
EIT / AT	electromagnetically induced transparency / Autler–Townes
HFSS / FEM	Ansys HFSS / finite-element method
PCB / FSS	printed circuit board / frequency-selective surface
LCP / RCP	left / right circular polarisation
VNA / SDR	vector network analyser / software-defined radio

I. INTRODUCTION

A microwave can reach a receiver with the correct frequency and field strength yet carry an unknown polarisation. Its two transverse electric-field components are described by a relative amplitude and phase; together these determine linear, elliptical and circular states. A scalar RF amplitude measurement therefore leaves important information unresolved. Polarisation-aware sensing is relevant to emitter characterisation, electromagnetic metrology, antenna alignment and propagation analysis.

Rydberg atoms provide a compelling optical interface to microwave fields. Probe and coupling lasers establish electromagnetically induced transparency (EIT) in a rubidium vapour cell, while an applied RF field perturbs the coupled high-lying atomic states. Spectral splitting or optical transmission changes can reveal microwave coupling without requiring a conventional electrical receiving element inside the vapour. The thesis laboratory used a 780-nm probe and 480-nm coupling laser; the experimental platform and its proposed metasurface front end are shown in Fig. 1. The sensor supplies the field-sensitive readout. The polarisation-selective metasurface supplies a configurable electromagnetic measurement basis.

The central engineering question is whether one planar RF structure can produce enough independent responses to recover the four transverse Stokes parameters at a fixed receiving axis. A practical answer requires more than a visually asymmetric unit cell: the six state-dependent Jones responses must form a full-rank, adequately conditioned Stokes instrument. We develop a compact two-sided copper geometry on one dielectric substrate, evaluate its complex HFSS response, optimise the trace geometry over L-band and test a finite receiving region. Experimental linear-polarisation measurements provide an independent laboratory foundation.

The principal contribution is a reproducible *HFSS-derived full-Stokes electromagnetic analyser*, accompanied by a calibrated mathematical inversion and a bounded geometry-optimisation method. The project does not assert that a commercial PIN-diode array or complete rubidium full-Stokes receiver was fabricated. The exact differences between periodic transmission, finite-panel receiving fields and optical atomic readout are maintained throughout.

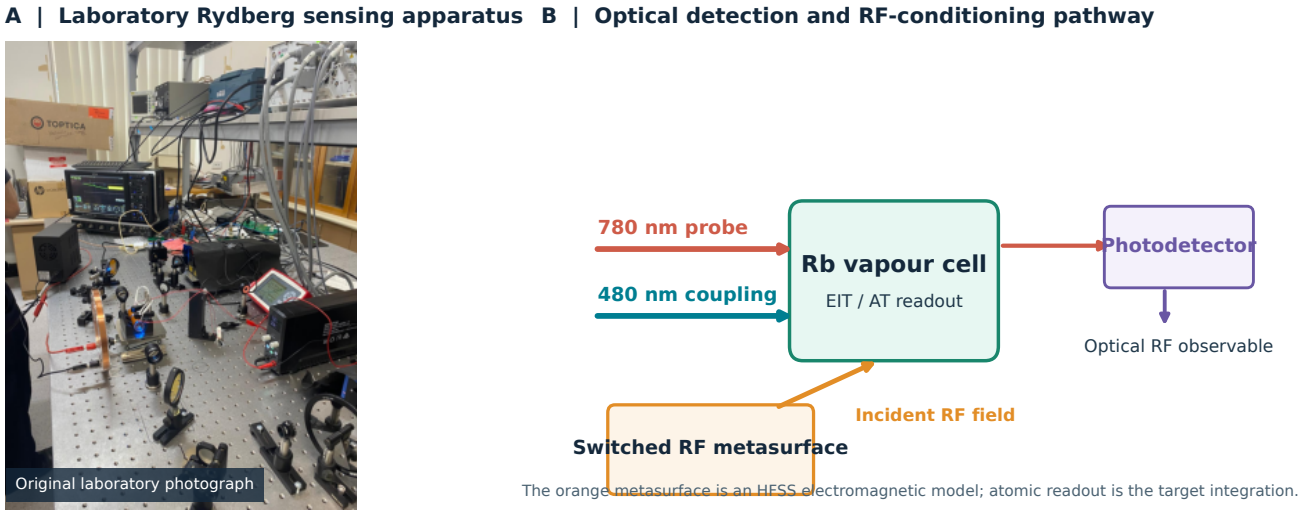


FIG. 1. Laboratory rubidium receiver and proposed polarisation-conditioning interface. Photograph from the author's original poster material[16]; connections are schematic.

II. RYDBERG DETECTION AND POLARISATION FOUNDATIONS

A. From a microwave electric field to an optical observable

In a ladder-type Rydberg-EIT measurement, a probe beam drives the rubidium ground-to-intermediate transition and a coupling beam connects the intermediate state to a Rydberg state. A microwave field couples neighbouring Rydberg levels and modifies the susceptibility experienced by the optical probe. In the appropriate near-resonant, resolved Autler–Townes limit, the RF coupling rate is

$$\Omega_{\mathrm{RF}} = \frac{|\boldsymbol{d}_{ab} \cdot \boldsymbol{E}_{\mathrm{RF}}|}{\hbar}, \tag{1}$$

where $\boldsymbol{d}_{ab}$ is the relevant transition dipole moment. The corresponding AT separation can be related to the RF field amplitude after accounting for the atomic transition, detuning and spectral resolution[1,2]. Figure 2 illustrates the optical coupling; its lineshapes are schematic, not measured spectra. An actual rubidium spectrum also depends on Doppler averaging, optical powers and linewidths. These atomic parameters are relevant when connecting an electromagnetic simulation to a real sensor.

The dot product in Eq. (1) also explains the polarisation ambiguity. A single dipole projection constrains a field component but does not determine both transverse amplitudes and their phase. Complete transverse polarimetry instead requires independent intensity-sensitive projections that sample the Stokes information. This thesis places those projections in the RF structure while leaving the subsequent atomic transfer function as an experimentally calibratable interface.

B. Jones, Stokes and Poincaré representations

Write the incident Jones vector as $\boldsymbol{j} = (E_x, E_y)^{\mathsf{T}}$. The Stokes four-vector $\boldsymbol{S} = (S_0, S_1, S_2, S_3)^{\mathsf{T}}$ contains total intensity, horizontal–vertical contrast, diagonal–antidiagonal contrast and circular handedness. The sign convention used throughout is $S_3 = 2\,\mathrm{Im}(E_x E_y^*)$. Purely polarised states satisfy $S_1^2 + S_2^2 + S_3^2 = S_0^2$ and occupy the Poincaré sphere. The opposite circular states lie at its poles; all ideal linear states lie on the equator. The detailed operator derivation and convention check appear in Appendix B.

A rotated ideal linear analyser can access amplitude orientation but does not independently determine circular handedness. A multi-state RF surface must therefore respond to phase quadrature as well as linear-amplitude contrast. In the present study, two anisotropic copper faces and four configurable conductor gaps alter the complex transmitted Jones response. Six selected configurations are sufficient to form a full-rank Stokes observation matrix in HFSS at the reported operating points.

C. What the optical receiver measures in practice

The apparatus brings together three very different forms of control: atomic frequency selection, optical preparation and microwave-field excitation. The 780-nm beam interrogates an optical transition in rubidium. The 480-nm coupling beam connects the intermediate state to a highly excited Rydberg state and establishes a narrow transparency feature under suitable detuning and coherence conditions. The incoming microwave couples two nearby Rydberg states, changing the dressed-state spectrum of the optically interrogated medium. A photodetec-

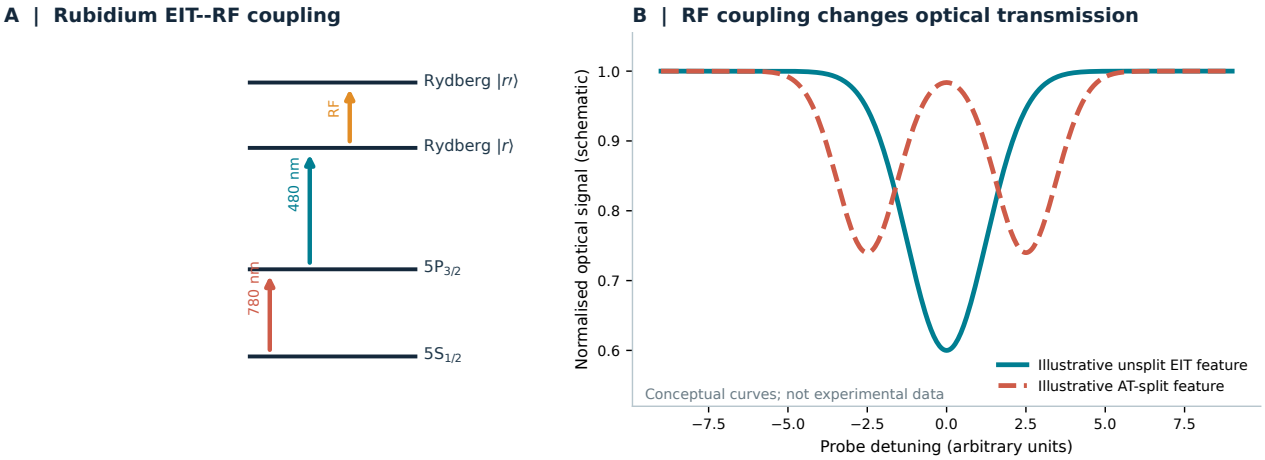


FIG. 2. Rydberg EIT and RF coupling principle. The conceptual lineshapes illustrate the optical consequence of microwave coupling; no artificial measurements are implied.

tor converts transmitted optical power into an electrical record from which resonance position, absorption contrast, frequency splitting or other calibrated features can be extracted. Figure 1 locates each function within the laboratory architecture.

These stages matter because the spectral feature is not the microwave electric field itself. In a resolved AT measurement, the transition dipole and frequency splitting provide a connection to field amplitude. Under weak coupling, unresolved spectral features, substantial optical power broadening or changes in field polarisation, the relationship is more involved. The optical detector records an ensemble response from many atoms across the illuminated volume and thermal-velocity distribution. Consequently, an accurate electromagnetic front-end model and an accurate optical readout model solve distinct portions of the engineering problem. This distinction underlies the validation boundary used throughout the thesis.

The experimentally available instrument demonstrates the optical coupling concept and supports conventional RF measurements; the metasurface models contribute a configurable set of incident-field transformations. With sequential switching, an unknown incident Jones vector remains unchanged between nominal measurement states while the RF structure alters the field sampled in the vapour region. Each state supplies one calibrated scalar response. A full-Stokes reconstruction becomes possible when those state responses contain independent linear, diagonal and circular information. The required field coherence and stabilisation assumptions should be assessed when translating the design to hardware, but they do not prevent numerical investigation of its electromagnetic response.

D. Why circular polarisation requires a phase-sensitive basis

Consider two incident fields $\boldsymbol{j}_+ = (1, \mathrm{i})^\mathsf{T}/\sqrt{2}$ and $\boldsymbol{j}_- = (1, -\mathrm{i})^\mathsf{T}/\sqrt{2}$. Both have the same magnitude in the orthogonal linear axes; their difference is the sign of the relative quadrature. Any receiving operator that depends only on separate noninterfering $|E_x|^2$ and $|E_y|^2$ terms gives identical observations for both fields. The distinction requires a nonzero cross term involving the complex product $E_x^* E_y$. The importance of phase-aware complex scattering data follows immediately from the expansion of the quadratic field magnitude: the cross term changes sign with circular handedness when the receiving state supplies an imaginary interference coefficient.

On the Poincaré sphere, the normalized input field is represented by (s_1, s_2, s_3). A sequence of idealised analyser projections creates a measurement vector $\boldsymbol{m} = A\boldsymbol{S}$. If the rows of A all lie in the same subspace, particular polarisation changes can occur without affecting the measurements. If the rows span all four Stokes coordinates, a unique algebraic inverse exists within the assumed measurement law. The spread of row directions also matters. An instrument with four technically nonzero singular values can be extremely sensitive to small calibration or field changes when its smallest singular value is tiny. That is why the geometric optimisation here uses conditioning, not just numerical rank or amplitude extinction.

E. Selection of the method and comparison with alternatives

The project began with a receiver-centred polarisation hypothesis. A controlled magnetic field could in principle select a quantisation axis and alter the relative optical response to π and $\sigma^\pm$ RF coupling. The interim experiment calibrated the coil, considered the influence of optical

linewidth and demonstrated a stable RF polarisation reference, but the available magnetic-bias scans did not yield a sufficiently repeatable polarisation estimate. This pointed to an engineering decision: rather than depend on resolving several weak atomic features, establish a separate set of polarisation-sensitive transformations in the RF path and treat the atom as the subsequent field sensor.

There are several possible routes to that capability. Spatially resolved polarimeters use multiple detectors or output diffraction channels. Coherent heterodyne systems compare an unknown field with several reference axes and can directly recover phase. A mechanically rotated polariser sequentially changes the linear measurement basis but adds a moving part and does not automatically supply circular information. Resonant anisotropic sheets can be compact, yet a single fixed geometry may preferentially distinguish only two states. Electronically reconfigurable structures combine sequential acquisition with a stationary receiving location. For the laboratory apparatus, the last approach offers a natural way to explore full-Stokes information with only one proposed PCB placed in front of the optical receiving region.

The method chosen for this thesis accordingly has four nested levels: a measurement reference obtained from physical RF polarisation experiments; a full-wave finite-element solution for a periodic parameterised cell; a mathematical inverse problem derived from the simulated complex fields; and a distinct finite-array simulation that samples an actual receiving region. Each level answers a separate question. The experiment demonstrates hardware orientation sensitivity. The periodic model establishes that a design can in principle supply a well-conditioned set of polarisation measurements. The algebra verifies that the inverse calculation matches the model. The finite array examines how finite aperture, field variation and observation distance affect the response. These levels are deliberately not counted as equivalent experimental validations.

F. Laboratory architecture and measurement chain

The underlying Rydberg sensing arrangement is a room-temperature rubidium system interrogated by narrow-linewidth optical sources. The red probe beam and blue coupling beam overlap in the vapour cell; their relative alignment determines the number of atoms contributing to the optical response and the shape of the EIT feature. The transmitted probe reaches a photodetector, whose electrical output is observed or analysed using external acquisition hardware. Microwave excitation is delivered from a controllable source through the laboratory RF path and reaches the optical overlap region. The relevant measurement is not the total microwave energy radiated into the room, but the field component coupling the chosen Rydberg transition within the illuminated atoms.

A measurement system built this way separates the atomic reference from the microwave coupling geometry. Atomic level differences and transition dipoles define the frequency and amplitude connection, while optics, thermal conditions and beam geometry determine whether that connection produces a clean, resolvable laboratory feature. In the actual experiment the sources and detector possess finite bandwidths and readout noise. Those matters were significant during the interim measurements, especially for magnetically biased sweeps that did not yield a repeatable additional analyser axis. The present electromagnetic modelling instead holds the receiving functional fixed so that the geometric contribution to polarisation information can be investigated before an optical reconstruction algorithm is calibrated.

The project equipment and assembly were documented in the earlier interim report and the poster presentation[15,16]. These records provide the source for the experimental architecture and orientation-response data used here. The field deployable receiver and desktop digital-twin software represent parallel systems-engineering activities, intended to improve portability and numerical understanding, but they do not modify the electromagnetic instrument matrices presented in this thesis. Their inclusion as engineering context is appropriate for an undergraduate research report; their validation is not silently borrowed to support metasurface polarimetric claims.

G. A measurement example: why amplitude alone is insufficient

Suppose an unknown field has transverse components $E_x = E_0 \cos\alpha$ and $E_y = E_0 \sin\alpha\, e^{\mathrm{i}\delta}$. The parameter α controls relative component amplitudes while δ controls their phase. At $\delta = 0$ the field is linearly polarised; at equal component amplitudes and $\delta = \pm\pi/2$ the field is circular. General combinations are elliptical. A power detector measuring only $|E_x|^2$ depends on α but is independent of δ. It cannot distinguish two equal-amplitude states whose phase difference has the opposite sign. Rotating an ordinary ideal linear analyser changes which real projection is seen, but a full phase-sensitive basis is required to recover the remaining circular coordinate.

A switched metamaterial does not need to create six new electromagnetic wave sources to solve this inverse problem. It can apply six distinct linear transformations to the same unknown Jones vector. If the receiver output for state k is $m_k = |t_{kx}E_x + t_{ky}E_y|^2$, the expansion contains squared terms in E_x and E_y as well as the complex interference term $t^*_{kx}t_{ky}E^*_xE_y$. It is that cross term which introduces sensitivity to relative phase. By collecting m_k over several configurations and calibrating each t_k from complex HFSS data, one can build a matrix whose columns encode all Stokes components. The equations are exact for the stipulated electromagnetic quadratic observation model. A future atomic experiment must establish that its measured optical feature can be mapped

onto that functional.

III. RELATED WORK AND RESEARCH OPPORTUNITY

A. Atomic microwave field sensing

Rydberg-based electrometry established optically interrogated RF field measurements through EIT and related spectroscopic observables[1,2]. Contemporary atomic receivers have advanced toward frequency, phase and vector information. Elgee *et al.*[8] demonstrated three-dimensional atomic RF polarimetry with orthogonal coherent local oscillators, providing a particularly relevant experimental comparator. Yin *et al.*[9] examined polarisation-sensitive atomic mixing. These studies establish that atomic vector sensing is feasible. Their coherent reference architectures motivate a complementary question: can a simple passive or switchable electromagnetic front end supply several calibrated projections to a receiver with a fixed measurement axis?

The laboratory's broader research programme supplies practical context, including room-temperature broadband rubidium sensing, RF mixing studies and temperature-dependent optical receiver response[14]. Those investigations explain why receiver architecture, EIT readout and field overlap matter. In this thesis, their existing data are cited as background; the claimed metasurface performance comes from the student's separate HFSS simulation records.

B. Polarimetric metasurfaces and microwave switching

Instrument-matrix polarimetry is established in classical optics[5–7]. Optical metasurfaces achieve compact full-Stokes analysis through engineered anisotropy and spatially separated polarisation responses[3,4,10]. At radio and microwave frequencies, patterned conductors, frequency-selective surfaces and electronic switches provide comparable degrees of electromagnetic control, although their wavelengths and material physics differ substantially[11,12].

The most direct competing category is reconfigurable microwave polarisation detection. Chen *et al.*[13] report an electronically configurable microwave architecture that can generate and detect varied incident polarisation states. Our contribution is therefore *not* a claim to the first full-Stokes metasurface or the first programmable polarimeter. It is the design and quantified HFSS optimisation of a compact, single-substrate, two-sided, six-measurement sequential analyser intended for a fixed receiving axis near an atomic vapour cell.

C. Research opportunity

A compact Rydberg receiver can benefit from a front end that delivers distinguishable electromagnetic responses without introducing multiple physical measurement axes into the sensing head. The research gap addressed here is an *optimised RF measurement basis*: geometry should be selected to improve the rank, conditioning and usable frequency span of the resulting Stokes instrument, not only its transmission or circular dichroism. The analysis must retain the complex Jones coefficients, since amplitude-only responses cannot identify relative transverse phase.

IV. RESEARCH OBJECTIVES AND DESIGN STRATEGY

The final research objective was to demonstrate a practical design route for transverse full-Stokes polarisation information using an L-band, single-substrate RF metasurface intended for Rydberg sensing. Three measurable tasks define success. First, laboratory measurements establish orientation-dependent linear RF response. Second, HFSS calculates a set of polarisation-sensitive Jones transmission operators from the same copper architecture under six conductor configurations. Third, constrained optimisation improves the sampled operating region while retaining full-rank Stokes observability. A finite 3×3 aperture is then simulated to examine the receiving-volume field response.

A physical implementation would replace each simulated ideal conducting bridge with an appropriately biased RF PIN-diode circuit. Here four proposed diode locations correspond to binary conductor paths. The study fixes a 24 mm unit-cell period, a 1.6 mm substrate and separate front/rear copper traces. The nominal projected manufacturing array would contain 7×7 cells within a 178 mm square. HFSS provides electromagnetic predictions for the ideal-bridge model; the actual PIN parasitics, bias network and Gerber release are subsequent integration tasks.

V. ELECTROMAGNETIC MODELLING AND VERIFICATION

A. Complex-field receiving model

For every switch configuration k, two phase-coherent orthogonal plane-wave excitations in HFSS produce a 2×2 complex Floquet transmission matrix $J_k(f)$. A fixed receiving-axis projection of J_k defines a quadratic electromagnetic observable. Its exact Jones-to-Stokes operator is derived in Appendix B; six such rows form the instrument $A(f)$, whose numerical rank determines whether all four Stokes components are accessible. The condition number of the row-normalised instrument gauges the independence of the measurement states. This is a specified HFSS receiving proxy; it is not an EIT spectrum or an experimentally measured atomic calibration.

TABLE I. Reference measurement architectures and the role of the proposed RF analyser.

Approach	Polarisation basis	Principal strength	Use in this study
Rydberg heterodyne[8]	Three coherent RF axes	Atomic vector measurement	Prior-work comparator
Optical metasurface[3]	Independent optical channels	Compact Stokes projection	Inversion precedent
Programmable RF surface[13]	Tunable electronic response	Reconfigurable microwave basis	Device comparator
This thesis	Six sequential switch states	Single-substrate RF encoding	HFSS design contribution

B. Geometry parametrisation and electromagnetic optimisation

The copper geometry uses front and rear asymmetric split-loop traces. Eleven continuous parameters describe front/rear in-plane scale, both trace widths, rear orientation, rear scale, conductor-gap length and two dimensionless gap positions. The script applies explicit width, gap and cell-clearance bounds before HFSS execution. Thirty-six candidates were evaluated using six switch states and a 41-frequency interpolating sweep from 1.0 to 2.0 GHz in 25-MHz steps. The objective favours adjacent sampled frequencies satisfying full rank, study-specific conditioning and minimum receiving-response thresholds. The reference, bandwidth-optimised and high-band alternative geometries retain an identical substrate and cell period.

C. Independent-frequency and finite-array calculations

Discovery sweeps use one frequency-adapted mesh and interpolated Jones responses; selected frequencies were subsequently resolved with independently adapted HFSS meshes. The strict dual-mesh comparison evaluates both the complex instrument and its reconstructed state transfer. We retain numerical screening thresholds of rank four, normalised condition number $\kappa_2 \leq 35$ and a 5% intermesh matrix agreement test, but never equate an interpolated sampled span with a guaranteed continuous bandwidth.

A separate finite-array study places nine cells on an 82 mm square dielectric panel, with radiation boundaries and normal-incidence x and y illumination. The resulting total electric fields are sampled over a 70 mm×50 mm receiving region at planes 5, 10, 20 and 50 mm behind the surface. The finite receiver matrix is derived from these complex field maps. Field-export provenance was checked using $\boldsymbol{E}_{\text{total}} = \boldsymbol{E}_{\text{incident}} + \boldsymbol{E}_{\text{scattered}}$. Mesh sensitivity is retained in the interpretation of the final results.

D. Preprocessing, simulation and data-governance workflow

The HFSS workflow begins with a geometry parameter file rather than manual adjustment of screenshots. It generates copper polygons for the front and rear faces, separates open and conducting gaps according to the selected state, verifies the unit-cell clearance and rejects invalid polygon fragments. The same parameter set is reused across the frequency screen and the selected-frequency verification runs. Fixed topology and fixed cell period permit a meaningful comparison of the sampled L-band performance as widths, trace scales, angles and gap positions change. All design iterations are recorded as parameter vectors, numerical model identifiers and calculated objective values.

The Floquet periodic model uses two independent transverse incident excitations for every switched state. Linear superposition yields a two-column Jones transmission matrix, which must be stored in complex rather than magnitude-only form. Its off-diagonal terms encode polarisation mixing, while the differences between opposite phase-quadrature inputs are determined by the interference terms of the full receiving operator. These output coefficients are converted to Stokes rows using a documented time-harmonic and circular-handedness convention. A failure to preserve those conventions could reverse the nominal LCP/RCP labels even if all magnitude plots looked unchanged. This is an important reproducibility control in any polarimetric microwave simulation.

The computational campaign distinguishes screening from validation. An interpolating frequency sweep efficiently reveals promising regions and permits a larger number of geometries to be compared within the available licence. Interpolation does not give a separately adapted tetrahedral mesh at every 25-MHz point. Therefore an independent-frequency HFSS solve is required to assess actual response stability at a selected point. The report presents those two layers of numerical evidence explicitly. It also archives negative or incomplete adaptive-solver statuses; a solver command returning without a fatal error is not sufficient evidence that a field of interest is stable under refinement.

Finite-plane-wave modelling introduces further considerations. Unlike a Floquet array, the finite panel has open lateral edges and a surrounding electromagnetic domain, so its fields include diffraction and spatial variation. In HFSS the calculation may separately export the incident field, scattered field and their total. The receiving-volume observable must use the physical total field rather than an arbitrary scattered-field export. The V4.13 field provenance experiment established that distinction through direct complex-vector addition. Field values from two incident orthogonal bases can then be combined coherently, allowing an explicit spatial receiving operator to be evaluated for arbitrary input polarisation without rerunning every test state in HFSS.

The study's data archive has been structured to avoid a common weakness of visually impressive simulation papers: a published graph should not be the only surviving representation of a scientific result. The necessary reproducibility files include the exact copper parameters, state mapping, solver settings, complex frequency data, figure-making scripts, mesh convergence records and the generated Stokes instruments. Raw native projects remain larger than the compact return archive and should be preserved alongside it. This separation permits the final diagrams, heatmaps and tables to be regenerated without guessing numerical values from pixels in an earlier report.

E. HFSS solution setup and quality assurance

A repeatable microwave simulation must specify both what it solves and what it extracts. For the periodic structure the Floquet ports excite two orthogonal transverse polarisations while master/slave or the equivalent periodic constraints model a repeating unit cell. The physical material stack consists of a dielectric sheet with patterned conductors on opposing faces. The geometry is normalised to a 24-mm cell and the relative phase between the two incident basis excitations is preserved. The complex transmitted modal coefficients are read from the same port convention for all six switching states and frequencies. This matters because a sign inversion, port-direction reversal or swapped linear basis can change inferred Stokes handedness without making a magnitude-only S-parameter spectrum look suspicious.

A separate empty-cell or empty-board reference provides a consistent electromagnetic baseline. It is not a substitute for an independent measurement axis: it corrects or documents common propagation and substrate effects while the switched states supply the polarimetric diversity. The instrument rows are derived from the complex Jones matrices, and calibration data are retained with their HFSS setup identifiers. The optimisation compares candidate structures at equal unit-cell pitch and material-stack assumptions; otherwise the apparent gain of a geometry change could be caused by a different aperture or reference setup.

The selected-frequency verification is intentionally more expensive than the discovery sweep. A discrete adapted HFSS solution examines the response at one particular frequency with its own local mesh; two such solutions, using controlled refinement policies, allow the complex Jones coefficient vector to be compared. A small difference between the resulting instrument matrices supports numerical repeatability. A different criterion evaluates what happens if an instrument calculated from one mesh is used to invert observations generated by another. These two tests answer different questions: matrix change measures electromagnetic coefficient stability, whereas cross-model inversion assesses how that change is amplified by the inverse operator. The latter is particularly sensitive when a measurement basis is nearly singular.

The FEM meshes are not calibrated physical measurements; they are numerical approximations that become more useful when the relevant outputs stabilise under independent refinement. Earlier simulations in this project sometimes reached the maximum adaptive passes without meeting their actual convergence targets. The later campaigns therefore captured HFSS native convergence files and compared both the declared setting and the criterion actually used by the solver. For the finite plane-wave problem, the relevant adaptive criterion was the magnetic-energy change rather than the port-based Delta-S quantity. The distinction explains why a solver can report completion while local near-field predictions remain insufficiently converged for a numerical accuracy claim.

The Student edition's 64,000-volume-element limit imposed a practical constraint on the finite mesh campaign. This should not be confused with a fundamental electromagnetic limit of the proposed surface. The highest-cost geometries require more local refinement around current concentrations and conductor gaps than the licence permits. The thesis therefore distinguishes a completed finite-field data set from a proven mesh-independent field response. The former is still valuable, because it reveals full-rank model behaviour and observation-plane dependence; the latter requires either more targeted refinement or a solver environment with a higher mesh allowance.

F. Instrument construction and physicality checks

For a complex Jones row $\boldsymbol{t}_k = (t_{kx}, t_{ky})$ representing a selected receiving axis, define the non-negative Hermitian matrix $Q_k = \boldsymbol{t}_k^{\dagger}\boldsymbol{t}_k$. The expected quadratic output is $m_k = \boldsymbol{j}^{\dagger}Q_k\boldsymbol{j}$. Since Q_k is positive semidefinite, its eigenvalues cannot be negative. This basic constraint is a useful implementation check: a fitted analyser row that predicts negative power for a physically admissible Stokes vector is inconsistent with the specified receiving model or has been processed with incompatible conventions.

Each Hermitian 2×2 response can be represented by a four-element real Stokes measurement row. The full instrument $A \in \mathbb{R}^{6\times4}$ therefore maps the four incident Stokes coordinates into six real model observations. A rank-four instrument admits a least-squares inverse even though it contains redundant measurements. If the rows are too similar, the smallest singular value becomes small and the inverse is poorly conditioned. The numerical pipeline reports all four singular values and the normalized condition number, since rank alone is an inadequate measure of polarisation discrimination quality. The six measurements also provide an opportunity to check internal consistency of the linear model because the observed response vector occupies the column space of A in the ideal calculation.

The full derivation in Appendix B generalises the operator from a single projected Floquet coefficient to a

spatially weighted receiving region. That generalisation is central to the finite HFSS study: different incident polarisations produce different complex total-field distributions, and the receiver effectively weights the two orthogonal basis fields according to its position and extent. An intensity-like integration over a specified region yields a physically non-negative quadratic form and an explicit measurement row. Changing that region changes the actual measurement definition. The board and the atomic optical mode consequently need to be optimised together in any future complete receiver design.

VI. MEASURED LINEAR-POLARISATION RESPONSE

A. Theory and experimental method

An ideal linear analyser transmits the electric-field component aligned with its axis. For incident linear polarisation at relative angle θ, the familiar intensity law is $I(\theta) \propto \cos^2\theta$; real RF polarisers exhibit finite extinction, insertion loss and geometry-dependent departures from this reference. The laboratory campaign used a linear source, a metal-rod polarising structure and a receiving arrangement near 2 GHz to quantify orientation-dependent attenuation. A separate interim VNA campaign characterised the steel-rod polariser and the magnetic-bias coil. These are different source datasets and are reported as such.

B. Measured results

The later archived point readings were -8.2 dBm without the front-end polariser, -13.9 dBm in the preferred orientation, -15.1 dBm at nominal 45° and -23.6 dBm in the rejected orientation. The preferred-to-rejected contrast is **9.7 dB**. The independent June interim report recorded approximately 14–15 dB pass/reject VNA contrast for the custom steel-rod configuration[15]. Figure 3 retains these contexts and uses the original thesis poster plot without inventing unreported samples. The experimental trend is consistent with substantial linear-orientation selectivity.

The reported preliminary normalised equatorial Stokes estimate was $s_1 \simeq 0.806$, $s_2 \simeq 0.370$. Its projected azimuth is $\frac{1}{2}\mathrm{atan2}(s_2, s_1) \simeq 12.3°$; this estimate contains no independently recovered circular coordinate s_3. The measurement therefore motivates a state-dependent RF analyser that adds a phase-sensitive response while keeping the existing Rydberg platform conceptually simple. The coil calibration $B_0(I) \simeq 2.302I + 0.028$ mT and its reported $R^2 = 0.981$ document an additional experimental subsystem[15]; they are not used to manufacture a polarisation result.

C. Experimental interpretation and measurement limits

The four later received-power readings support two meaningful performance quantities. The 9.7-dB preferred-to-rejected difference quantifies angular discrimination between two configurations of the measured polariser. By contrast, the difference of 5.7 dB between the unscreened path and the preferred orientation includes the presence of the structure and the changed test path. It would therefore be misleading to call both numbers extinction ratio. The raw dBm difference between preferred and rejected positions corresponds to a received-power ratio of approximately $10^{9.7/10} \approx 9.33$, under the usual matched-power interpretation. The ratio gives a useful sense of orientation selectivity without imposing an optical Malus-law fit on four heterogeneous RF point measurements.

The earlier steel-rod VNA campaign and later orientation-dependent receiving experiment should not be merged into one fitted series. Different antennas, references, sample geometry and receiver channels can shift absolute received power and apparent contrast. The reported 14–15-dB interim result is valuable independent validation that the hardware architecture could substantially discriminate orientation, whereas the later 9.7-dB result illustrates its response in the context of a different readout chain. Both can be included in a final engineering account when their apparatus and definitions are specified separately.

The reported partial linear Stokes values also have a precise limit of interpretation. An observed (s_1, s_2) point estimates a projection onto the equatorial plane but does not determine s_3. Many elliptic states share the same linear coordinates while differing in handedness content. The experiment therefore supplies a clear motivation for the phase-sensitive metasurface: it provides the missing physical category of measurement without requiring the experimental record to imply a full polarisation measurement that was not performed.

VII. RECONFIGURABLE TWO-FACE METASURFACE: DESIGN THEORY

A. Why two anisotropic layers are useful

A useful circular analyser must discriminate fields of equal transverse intensity but opposite relative phase. A single real-valued projection provides linear-amplitude discrimination, whereas coupling between differently oriented resonant currents can generate the complex off-diagonal Jones coefficients needed to weight S_3. In a compact printed surface, trace length contributes inductive loading, open gaps contribute capacitive loading and the two opposed copper faces couple electromagnetically through the substrate. Changing the conduction across selected gaps alters these interactions and consequently changes the Stokes analyser row.

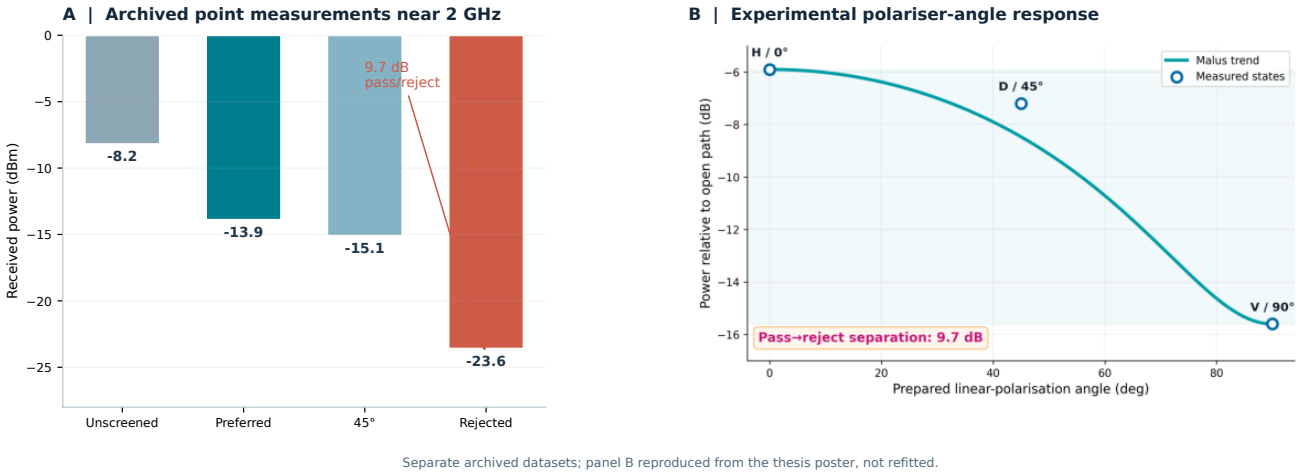


FIG. 3. Archived RF polariser measurements: discrete received powers and an original polarisation-angle response plot from the thesis poster.

The implemented geometry uses a split-loop-like copper path on each face of one dielectric sheet. The rear path is mirrored, rotated and scaled relative to the front. Four ideal conductor-switch sites are distributed across both faces; six switch combinations were selected for numerical inversion. Figure 4 presents the simulated 3×3 aperture, an enlarged cell and an equivalent-circuit interpretation. The circuit illustrates the physical design reasoning rather than a fitted lumped-element parameter extraction. PIN-diode symbols identify the proposed physical switching technology; HFSS calculated binary ideal conducting bridges.

B. Polarisation measurement rather than polarisation generation

The design objective is measurement-state diversity. It is not necessary that every switched structure emit an arbitrary target polarisation. Instead, its response to a fixed unknown incident field must provide an independent row of the common Stokes observation matrix. The tensor quantities responsible for coupling and circular sensitivity arise from the *complex* Jones response; consequently, simulated reflection or transmission magnitudes alone are insufficient for full-Stokes analysis. This distinction motivates the project-derived receiving operator, rank test and numerical condition-number objective.

C. Equivalent-circuit interpretation and switching-state diversity

The conductor path and interruption points can be interpreted through the stored-field physics of distributed microwave networks. Current along the elongated copper traces produces magnetic energy and an effective series inductive loading. Open gaps between conductor ends support local electric fields and capacitive displacement currents. Placing two different traces on opposite faces of a dielectric adds interlayer electric and magnetic coupling, including a phase delay associated with the geometry. The resulting array is not simply a conventional lumped L–C filter; rather, the equivalent elements explain why a changed physical conducting path can shift resonance, transmission phase and cross-polarisation conversion. The actual values of these reactive elements would require a dedicated model-order reduction or extracted network fit.

Each proposed PIN-diode position is a switch site between two conductor sections. The ideal HFSS model represents a closed state as an inserted conducting bridge and an open state as a gap. This construction is a controlled initial step because it isolates the influence of topology before manufacturer-specific diode parasitics and DC-bias traces are selected. A physical PIN implementation would require an ON-state series resistance and package inductance, an OFF-state junction capacitance, RF chokes or bias tees where appropriate and a bias routing topology that does not spoil the desired current distributions. The circuit inset in Fig. 4 deliberately separates the design interpretation from the simulation's ideal-switch assumption.

There are sixteen possible four-bit conductor states in principle. This work selects six masks that exhibit useful and distinct Jones projections. The state choice is a polarimetric measurement-basis decision rather than a digital modulation alphabet. The array is interrogated sequentially under the assumption that incident polarisation is stationary over an acquisition sequence, or that its evolution can be tracked by a sufficiently rapid controller. The important calibration output is therefore a row of the Stokes instrument for each selection. Equal transmitted power under one incident input does not imply equal utility: two states can respond almost identically to

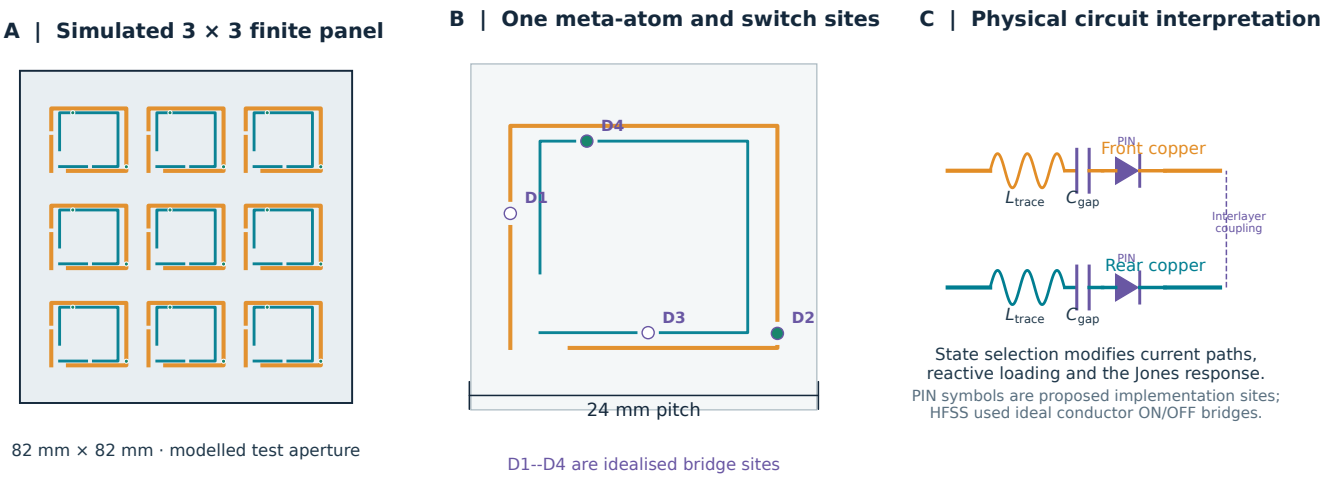


FIG. 4. Finite nine-cell model, enlarged two-face meta-atom and conceptual L–C–PIN switching interpretation. PIN circuits were not included in the ideal-bridge HFSS solves.

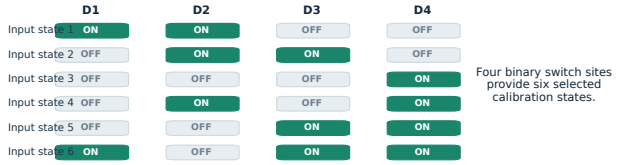


FIG. 5. Six sequential conducting configurations. Green denotes an ideal closed conductor bridge; binary states are labelled for proposed diode locations.

horizontal radiation yet differ markedly when the input includes circular quadrature.

The geometric constraints also govern the physical plausibility of the proposed PCB. The copper exists on two accessible external faces of a single dielectric sheet, and the high-level model uses a 1.6-mm board thickness. The chosen front and rear trace widths lie comfortably above the project's nominal minimum-feature target. Nonetheless, optical access through the full vapour-cell region, the substrate's frequency-dependent permittivity, copper plating, bias routing, drill/via requirements and diode footprint cannot be inferred from ideal-PEC field plots. The numerical model demonstrates an electromagnetic concept with parameter bounds; it is not a signed-off PCBWay fabrication package.

D. How the geometry parameters affect the measured response

The front and rear conductor shapes are deliberately related but not identical. Separate x and y scale parameters determine their projected lengths and the relative detuning of the orthogonal current paths. Trace widths influence the distributed inductance, conductor current distribution and gap-field concentration. The rear-layer rotation changes how the two anisotropic paths couple under incident fields aligned with the laboratory axes. The additional rear scale parameter adjusts their relative size, while the gap locations determine which portions of the resonant paths are interrupted when a switch opens. Altering a gap position may change the polarisation response even when the total copper area changes only slightly, because it moves the discontinuity relative to current antinodes and phase-sensitive coupling regions.

The parameters were varied within bounded ranges to preserve the nominal unit-cell footprint and conductor dimensions. The resulting optimization is not a search among completely unrelated shapes; it is a systematic experiment inside a controlled family of two-face split loops. This matters when interpreting the chosen design. The improved condition and sampled operating span can be attributed to changed geometrical relationships within one architecture, not to an uncontrolled change from one kind of antenna or substrate to another. At the same time, the limited search family places an honest boundary on the word *optimal*: the reported candidates are the best found among the evaluated parameter combinations and their stated screening objective, rather than a proof of globally optimal full-Stokes measurement.

The comparison can be understood in terms of an instrument basis on the Poincaré sphere. An idealised highly symmetric set of measurement projections is distributed so that no Stokes coordinate is observed much more weakly than the others. The present planar geometry cannot directly impose those ideal points at will; instead, HFSS supplies a physically constrained set of analyser rows for each switch pattern. Optimising geometric dimensions changes both the length and direction of those rows in Stokes space. The condition number measures this directional diversity after gain normalisation, while the minimum response metric prevents an appar-

TABLE II. Reader-facing design names and archived HFSS identifiers.

Design name	Archive code
Reference split-loop	C000
Bandwidth-optimised split-loop	C033
High-band alternative	C015

ently excellent condition number from being selected when one or more output channels are unusably weak. This joint optimisation gives the final design a more meaningful interpretation than a narrow resonance tune.

VIII. GEOMETRY OPTIMISATION AND FULL-STOKES SIMULATION RESULTS

A. Reference and two optimised structures

The reference split-loop provides the controlled electromagnetic baseline. A bounded 36-candidate search identified a bandwidth-optimised geometry as the leading solution and a high-band alternative as a useful comparator. All three preserve the same basic topology while modifying eleven dimensions and relative-layer parameters. Figure 6 shows the geometries at one common switch configuration, using a common coordinate system. The internal simulation identifiers are recorded once in Table II; reader-facing names are used elsewhere.

The primary optimised cell uses a front copper width of 1.400 mm, rear copper width of 1.1065 mm, rear rotation of 46.34° and nominal gap length 1.8867 mm. The front/rear trace scale factors and gap fractions are preserved with full precision in the electronic research archive. These parameters are HFSS model dimensions rather than released PCBWay Gerbers. The finite study uses a 24-mm cell pitch on an 82-mm panel; a nominal 178-mm 7-by-7 manufacturing outline is a separate projected arrangement.

B. Optimisation history and candidate selection

The reference design passes eight adjacent frequencies on the original 25-MHz discovery grid, spanning 1.825–2.000 GHz (175-MHz endpoint difference). Both optimised designs pass eleven sampled points, yielding a 250-MHz endpoint difference. The 42.9% span increase is a *sampled HFSS discovery metric*, established across 36 actual candidate HFSS campaigns. It does not prove that every frequency between the sampled endpoints or any untested frequency beyond 2 GHz satisfies the full-Stokes criterion. Figure 7 shows the complete candidate ranking and the trade-off between sampled span and matrix conditioning.

C. Optimisation metrics and what their improvement means

The optimisation objective is based on the receiving matrix rather than on an isolated S-parameter peak. First, an instrument must be rank four: the selected RF transformations cannot all project onto one linear subspace. Second, the rows must possess sufficient directional diversity, assessed through the condition number after row normalization. Third, the response should not vanish in one state, because a mathematically independent but arbitrarily weak measurement channel would be difficult to calibrate. A useful frequency requires all these criteria to be considered together. The screening thresholds are engineering decisions declared before evaluation, not universal definitions of a polarimeter's usable bandwidth.

With these conditions, the discovery score counts passing samples on a finite frequency grid. The reference geometry contributes eight contiguous 25-MHz sample points; the two optimised designs contribute eleven. The apparent advantage is observable in Fig. 7, which retains all 36 candidates rather than presenting only the winner. The distinction between the complete ranking and one final performance curve makes the optimisation credible: the reader can see the variation among designs, the selected objective and the possibility of several nearby solutions in geometry space. The high-band alternative is included for technical comparison because different parameter sets can perform similarly on the same scalar objective while changing the distribution of signal response and condition across frequency.

The independent adaptation stage provides a separate check. The attractive sampled performance around 1.925 and 1.950 GHz is supported by explicit HFSS adapted-frequency solutions, with both the condition numbers and intermesh matrix changes reported. Since these frequencies differ, the smaller of two condition numbers is not automatically the superior overall geometry. Selecting the final design should also consider the width and location of its passing frequency region, minimum receiving response and field behaviour in the intended finite aperture. The bandwidth-optimised design is retained as the principal subject here because it provides a useful balance and is the one carried through the finite-panel study.

One limitation of reporting a count-based sampled span is that no two sampled endpoints reveal the precise band edges. A hypothetical response can satisfy every measured vertex but dip below a criterion between two vertices if it contains a sufficiently narrow resonance. Consequently, the present 250-MHz value is best understood as a reproducible HFSS optimisation-screen measure. It should not be called an experimentally demonstrated continuous 250-MHz operating band. The independently adapted results at selected samples and the complete complex Jones archive are available to support a later, more densely verified band-edge study if that narrower claim becomes important.

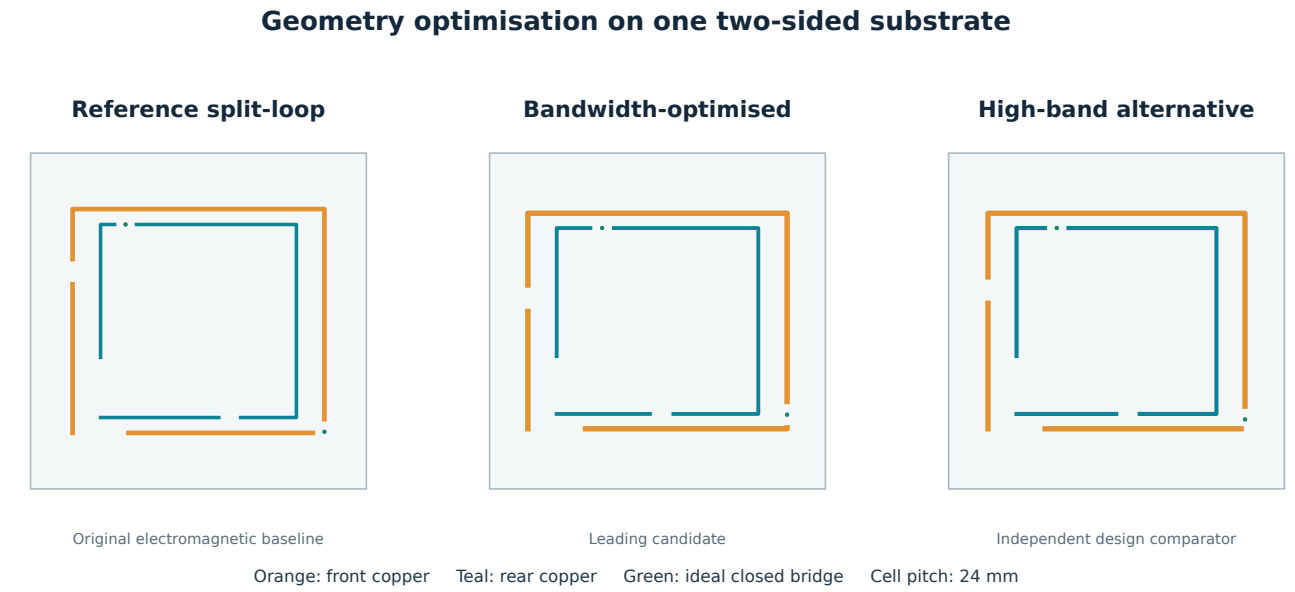


FIG. 6. The reference split-loop, bandwidth-optimised design and high-band comparator use one substrate with independently dimensioned front and rear copper traces.

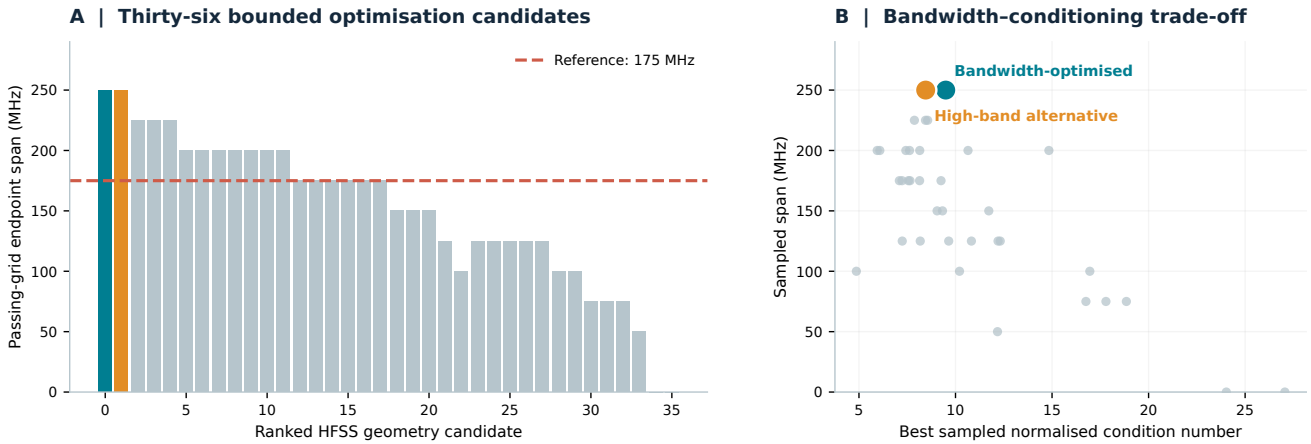


FIG. 7. HFSS optimisation across 36 geometry candidates. The dashed reference is the original 175-MHz sampled passing span.

D. Independent-frequency HFSS verification

A separate dual-mesh campaign independently adapted the six state-dependent HFSS solutions at selected frequencies. The bandwidth-optimised design retained full rank and reached normalised condition number 34.62 at 1.750 GHz, 9.53 at 1.925 GHz and 16.84 at 1.975 GHz. At 1.925 GHz its relative Stokes instrument-matrix change between mesh levels was 0.551%, while the deterministic calibration-transfer discrepancy on the defined numerical test set was 5.01%. The high-band alternative yielded a condition number of 8.64 at 1.950 GHz with 0.325% matrix change. These selected-frequency results provide stronger evidence than the discovery interpolation alone.

The reference design separately achieved a rank-four matrix at 1.850 GHz with normalised condition number 22.06 and 0.749% intermesh matrix difference. Its controlled-mesh response provides a stable baseline against which optimisation was evaluated. Because lower condition numbers at very different frequencies cannot be compared as an isolated performance improvement, the figure preserves frequency on the horizontal axis for both candidates.

E. Poincaré-sphere reconstruction and physical observability

The six native complex HFSS transmission responses define a 6×4 Stokes instrument. At 1.925 GHz the bandwidth-optimised model has rank four. We applied

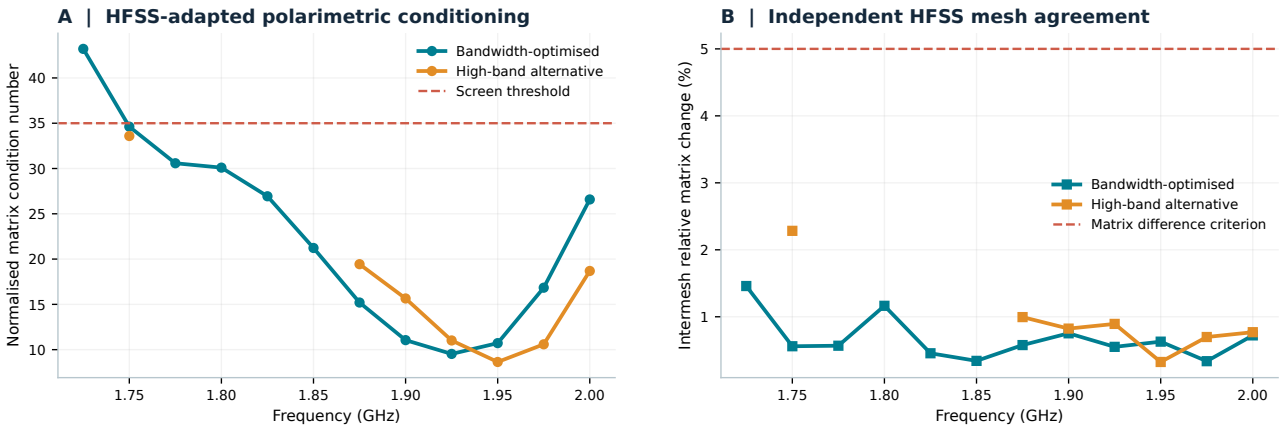


FIG. 8. Independently adapted HFSS frequencies: polarimetric condition number and matrix agreement for the optimised candidates.

its operator to 1,231 prescribed fully polarised Jones states, then used a singular-value-decomposition inverse to reconstruct their input Stokes parameters. The numerical residual is at the level of floating-point precision because the same HFSS-derived linear operator generates and inverts the test observations. These 1,231 positions are *algebraic tests*, not 1,231 separate full-wave HFSS solves or atomic optical measurements.

The six-state response map is scientifically more informative than a near-zero ideal inversion error: it shows why complementary measurements are required. The two circular states have identical S_0, S_1, S_2 but opposite S_3; their different predicted receiving responses demonstrate sensitivity to phase handedness. This is the electromagnetic capability needed before considering a future Rydberg optical transfer calibration.

IX. FINITE-PANEL RESPONSE AND NEAR-FIELD COUPLING

A. Finite electromagnetic experiment

The optimised split-loop was assembled into a 3×3, 82-by-82-mm finite electromagnetic model at 1.850 GHz, with plane-wave excitation under the two orthogonal input axes. Six ideal switch states give twelve HFSS patterned cases. The finite receiver was evaluated by integrating the squared transverse field over the sampled 70-by-50-mm receiving region, on planes 5, 10, 20 and 50 mm behind the panel. Incident, scattered and total field exports were separately identified, with total fields used for the final receiver calculations.

B. Six-state observability and distance dependence

The resulting receiving matrices are rank four under both completed HFSS mesh policies. On the 2.7-mm local copper mesh the normalised condition numbers at the four receiver planes are 24.72, 24.82, 59.81 and 209.15; on the 2.4-mm mesh they are 21.93, 30.17, 61.37 and 176.74. Figure 10 presents these complete results on a readable logarithmic scale. The strongest polarimetric separability occurs close to the surface, within the simulated 5–10-mm observation interval.

The trend is physically useful: finite-aperture near-fields and spatially varying polarisation generally relax toward a more uniform transmitted field as the observer moves away from the patterned surface. That propagation changes the diversity among switched RF states. As a result, a metasurface optimised using an infinite periodic far-field Jones matrix is not automatically optimal for a finite, spatially averaged atomic receiving volume.

C. Numerical scope of the finite-field result

The two finite meshes show materially different complex receiver fields for some states. At 5 mm the relative matrix difference between the completed finite policies is 6.18%, and deterministic cross-mesh reconstruction transfer can differ substantially. A later HFSS campaign tightened the actual incident-wave energy convergence criterion to $\Delta E = 0.02$; it converged the state-12/X and state-6/Y diagnostic cases but could not complete state-13/X within the Student edition's 64,000-element limit. Thus the completed finite result supports rank-four *model observability* and a useful receiver-distance study, whereas its exact near-field condition number is not independently field-mesh validated. A final 7-by-7 board and a complete atomic optical output are also outside the calculated dataset.

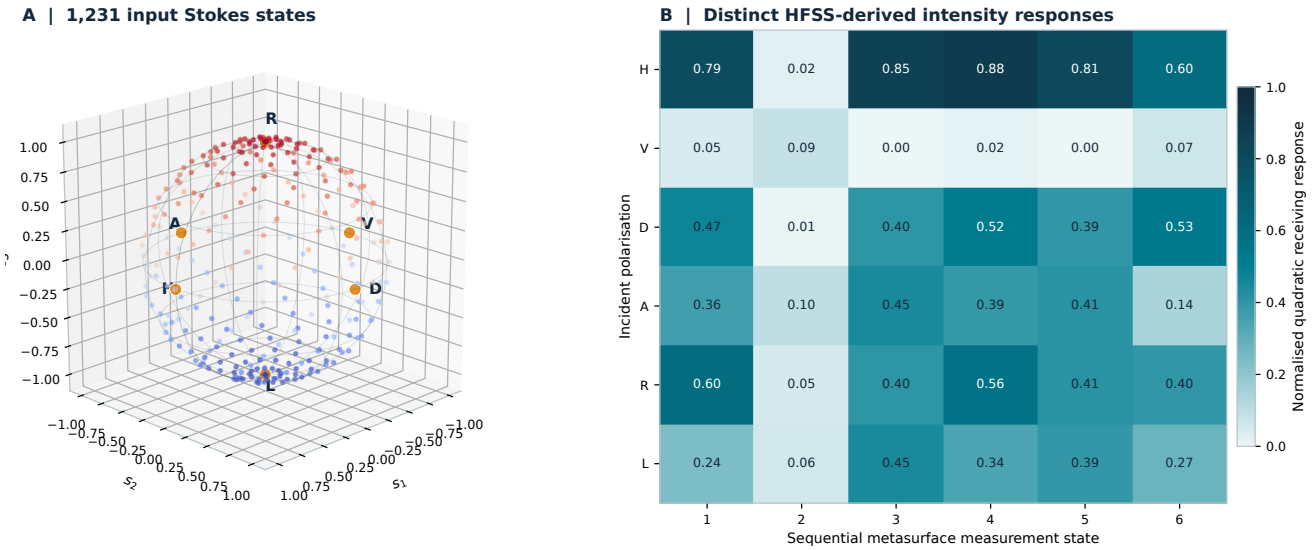


FIG. 9. HFSS-based Poincaré reconstruction and the six distinct responses to horizontal, vertical, diagonal, antidiagonal and circular input states.

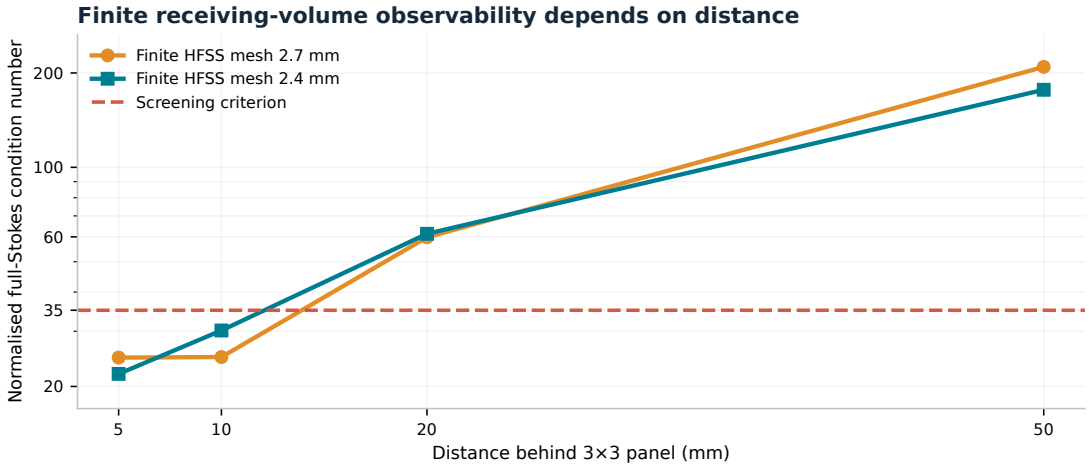


FIG. 10. Finite 3×3 receiving-matrix conditioning at four planes behind the PCB for two HFSS copper-mesh policies.

D. Finite receiving-volume physics and the observation-plane trade-off

Periodic HFSS Floquet coefficients describe transmission through an infinite array. A real printed panel terminates at its edges, and the resulting field can vary in direction, magnitude and phase across a nearby vapour-cell volume. The finite model thus addresses a different question from the periodic unit-cell response. A finite surface may retain polarisation-selective local fields even when its far-field transmitted beam is less strongly polarisation discriminating; conversely, a periodic model with strong cross-polar transmission does not by itself guarantee useful coupling to a particular finite detector area. The receiving-volume functional makes that distinction explicit.

The finite calculations used separate x- and y-polarised plane-wave bases. For each state, the complex transverse field at every sample point is a linear combination of these two solutions. Integrating its quadratic receiving-axis magnitude produces the instrument row appropriate to the declared observation plane. This construction is advantageous because a complete state sweep requires twelve physical HFSS cases rather than a separate electromagnetic solve for every incident point on the Poincaré sphere. The input-state enumeration used for reconstruction is consequently a transparent mathematical post-processing step.

The 5- and 10-mm planes yield normalized instrument conditions below the predefined threshold on both completed mesh policies. The 20- and 50-mm planes show

progressively poorer conditioning, even though the system remains rank four. This distinction is physically informative: algebraic observability and practical robustness are not identical. Near the surface, spatially structured fields can provide stronger differences between switching states; at larger distances, the state responses become closer and the condition number grows. Receiver placement must therefore be included among the RF design variables when the surface is intended to operate beside a small atomic sampling volume.

The remaining numerical qualification concerns the complex fields themselves. Finite-region field quantities can be sensitive to mesh refinement at sharp conductor edges and in strongly varying near-field zones. HFSS adaptive convergence in a global energy quantity cannot automatically certify every receiver-point field. The stringent follow-up calculations improved the native energy-convergence target and demonstrated that the Student mesh ceiling, rather than a completed high-fidelity three-mesh study, limited further validation. The article can report the complete finite matrix and its distance trend, but it must avoid claiming a fully mesh-independent 5-mm field map until the locally resolved complex responses agree. That boundary follows naturally from the data and does not erase the value of the finite receiving-volume study.

X. DISCUSSION: DESIGN INSIGHT AND ENGINEERING CONTRIBUTION

The research yields three connected findings. First, experimental linear discrimination demonstrates an RF measurement basis but leaves circular quadrature unresolved. Second, the two-face, four-switch geometry produces six complex HFSS analyser responses that form a rank-four Stokes operator, with the clearest periodic evidence around 1.925 GHz for the bandwidth-optimised design. Third, geometry optimisation improves the sampled operating span while the finite-panel study shows how the receiving-volume placement affects actual field diversity. These findings give a coherent engineering reason to design polarisation sensing jointly in the geometry, switching states and receiving functional.

The physical reason for using two layers is their additional electromagnetic coupling freedom. Angular asymmetry and reactive conductor gaps yield state-specific complex Jones coefficients; toggling a conducting bridge changes the current path and therefore the phase-sensitive response. The useful figure of merit is not only dichroic contrast under LCP/RCP illumination, but whether the complete set of switched responses spans S_0 through S_3 with acceptable conditioning. A single attractive handedness plot cannot establish full-Stokes retrieval.

The proposed RF structure remains a candidate Rydberg sensing interface rather than a fully integrated atomic polarimeter. In an experiment, the local microwave field would drive transition-specific Rabi frequencies, and optical overlap, detuning, hyperfine structure and readout nonlinearity would define the actual instrument rows. The present quadratic receiving proxy is clearly specified and reproducible, permitting useful RF engineering without assigning artificial optical spectra to HFSS. The next integration programme would calibrate the optical readout against known incident polarisation states, implement the PIN control network and characterise the final 178-mm panel.

The analytical contribution is an auditable workflow: complex Jones matrices are mapped to positive-semidefinite quadratic receiving operators, their Stokes rows are assembled, the condition and physical consistency of the inverse problem are checked, and frequency/aperture dependence is quantified. The algebraic identities derive from established electromagnetic and Stokes theory; our application-specific result is the coupling of these tests to a bounded, six-state, dual-face HFSS design programme. Their detailed derivations and test conventions are retained in Appendix B.

A. Implications for a sensor architect

The key design choice is where to place the polarimetric measurement basis. One strategy is to provide several atomic or microwave reference axes and solve for the field vector from coherent detector outputs. The present strategy places the configurable projections into the electromagnetic interface and leaves one intended receiving channel behind it. A future engineer choosing between these systems would consider the desired instantaneous bandwidth, the ability to switch states reproducibly, the cost of DC control, the permitted vapour-cell geometry and the difficulty of optical calibration. No single numerical condition number captures these system-level trade-offs.

The work also suggests a useful refinement to RF metasurface optimisation for sensing. Classical surface design may concentrate on transmission amplitude, reflection reduction or a chosen output polarisation. A polarimetric receiver should instead optimise the linear independence of the *measurement responses across configurations*, subject to sufficient signal strength. A geometry that maximises one circular-to-linear contrast need not be the best multi-state analyser. Conversely, a surface with moderate transmission discrimination in each individual state can form a strong inverse operator when its six projections are suitably distributed in Stokes space. The project's HFSS-to-Stokes implementation makes that distinction explicit.

The same mathematical approach is applicable beyond a Rydberg vapour cell whenever the receiving quantity can be expressed as a known positive-semidefinite quadratic function of the transverse electromagnetic field. Examples might include a calibrated power detector, a polarization-sensitive receiving aperture or spatially weighted antenna feed, provided the modelled spatial functional actually

matches the instrument. Changing the receiving functional changes the instrument rows, so its calibration is part of the physical measurement definition and should not be hidden in post-processing. This is one of the most general lessons from comparing the original periodic calculations with the finite sampling-volume response.

B. What the paper contributes beyond a final geometry

The design is a reproducible example of receiver-aware microwave metasurface engineering. A reader can start from the proposed two-face topology, use the documented parameter bounds to regenerate candidate geometries, simulate each state under two coherent input axes, assemble the complex Jones data and derive the corresponding Stokes receiving matrix. They can then apply a different spatial weighting appropriate to their own detector, or alter the optimisation threshold according to their application, while keeping the underlying mathematical structure unchanged. Providing the geometry, matrices and scripts makes the investigation more transferable than a plot of transmission coefficients alone.

The result should be compared with adjacent research honestly. An atomic heterodyne receiver and a switched RF metasurface solve the polarisation inverse problem through different measurement physics. Neither establishes the other's performance. An optical full-Stokes metasurface supplies valuable analysis ideas but uses a different wavelength, material stack and output arrangement. A published electronically reconfigurable microwave polarimeter is a close technology comparator and prevents overstatement of priority. The present work's distinguishing merit is therefore its particular compact two-face RF architecture, HFSS-backed full-rank sequential measurement model and direct focus on a small receiving region proposed for atomic sensing. In a future paper, matched-frequency, matched-aperture comparisons could quantify its advantage against those more mature systems.

The data also illuminate the limitations of the commonly used ideal-polariser analogy. Microwave conductor geometries are frequency-selective and may mix linear and circular components. Their Stokes rows can rotate rapidly near resonances, so a structure's performance at one frequency need not extrapolate across L-band. The finite aperture introduces a spatial dimension absent from an ideal infinite-array Jones matrix. These are engineering constraints rather than incidental complications: they determine whether a nominally observable inverse problem remains useful at the actual sensing location. The present design programme exposes those dependencies, making them available for other researchers to test against different receiving models.

XI. CONCLUSION

This thesis presents a reconfigurable L-band RF metasurface design pathway for transverse full-Stokes polarimetry motivated by rubidium Rydberg sensing. Laboratory linear-polariser measurements established a 9.7-dB preferred/rejected response near 2 GHz. A two-face split-loop structure with six selected conducting states subsequently produced full-rank HFSS Jones-to-Stokes instruments. Bounded optimisation of 36 candidate geometries increased the sampled passing span from 175 to 250 MHz. The selected bandwidth-optimised geometry achieved a normalised condition number of 9.53 at 1.925 GHz under independently adapted periodic HFSS solutions, and its six-state electromagnetic response supported deterministic Poincaré-sphere reconstruction.

The completed nine-cell finite-array model demonstrates full-rank receiving-field observability and highlights 5–10 mm as the stronger near-panel observation region. Finite near-field values remain mesh-sensitive, so the work is presented as a numerically characterised RF front end rather than a completed atomic full-Stokes instrument. The central outcome is a reproducible method for selecting and verifying polarisation-sensitive electromagnetic states on one substrate, with complete traceability from copper geometry through complex HFSS fields to Stokes reconstruction.

XII. FUTURE INTEGRATION AND RESEARCH HANDOVER

Practical continuation should prioritise a diode-populated PCB schematic and bias isolation, verified full-size finite-array field solutions, and an experimental optical transfer calibration between microwave receiving fields and measured rubidium EIT/AT observables. The project archive preserves the exact eleven geometry parameters, switching-state truth table, complex Jones matrices, HFSS convergence evidence, numerical Stokes arrays and editable scientific figures. Related desktop digital-twin and field-deployable sensor prototypes remain useful complementary engineering activities, but their functionality is separate from the full-Stokes performance claims of this study.

Appendix A: Scope and role of the supporting derivations

The derivations below are included because they specify the instrument actually calculated from native HFSS fields, the physical-admissibility constraints of that instrument, and reproducible inversion tests. They are not claimed as newly discovered universal laws, and an algebraic identity tested using the same fields is not counted as a new electromagnetic simulation. Their input arrays, conventions, and bounded domain are documented in the supplied numerical archive. Readers seeking only the

engineering conclusion can work from the main discussion and results; the appendix permits direct replication of the mathematical checks without inflating the thesis body.

Appendix B: Project-derived equations and analytical verification

1. Incident field convention and calibrated observables

Let a quasi-monochromatic incident field of fixed propagation direction have transverse Jones vector $\boldsymbol{j} = (a, b)^{\mathsf{T}}$. We use the Stokes convention

$$S_0 = |a|^2 + |b|^2, \qquad S_1 = |a|^2 - |b|^2, \tag{B1a}$$
$$S_2 = 2\Re(a^* b), \qquad S_3 = -2\Im(a^* b). \tag{B1b}$$

The sign of S_3 is a declared convention chosen to match the complex phasing of the HFSS post-processing code. Consequently, circular-handedness labels must be read together with the propagation and time-harmonic conventions. For fully polarized states $S_1^2 + S_2^2 + S_3^2 = S_0^2$; partially polarized states lie inside the normalized Poincaré sphere. Equation (B1) is established polarimetry, not an original theorem of this project.

At a given frequency f and switch state k, denote the HFSS complex electric-field component along the selected receiving axis by $u_k(\boldsymbol{r}; f)$ for an incident unit-amplitude x basis and $v_k(\boldsymbol{r}; f)$ for the orthogonal y basis. For a linear, time-invariant structure with the same material state and mesh, superposition gives

$$E_{x,k}(\boldsymbol{r}; f) = a\, u_k(\boldsymbol{r}; f) + b\, v_k(\boldsymbol{r}; f). \tag{B2}$$

The idealized scalar observation is a nonnegative, spatially weighted field magnitude,

$$m_k(f) = \int_{\mathcal{V}_\mathrm{r}} w(\boldsymbol{r}) \left| E_{x,k}(\boldsymbol{r}; f) \right|^2 \mathrm{d}V, \quad \int_{\mathcal{V}_\mathrm{r}} w \,\mathrm{d}V = 1, \tag{B3}$$

where $w \geq 0$ represents a specified receiving-volume sampling rule. In the principal numerical model, the available discrete field grid was uniformly weighted; a narrower central Gaussian weighting was considered separately. This is a modeled electromagnetic *intensity proxy*, not a calibrated Rydberg optical-signal law.

Define three complex-field integrals

$$P_k = \int w |u_k|^2 \,\mathrm{d}V, \quad Q_k = \int w |v_k|^2 \,\mathrm{d}V,$$
$$C_k = \int w u_k^* v_k \,\mathrm{d}V. \tag{B4}$$

Inserting Eq. (B2) into Eq. (B3), and using $a^* b = (S_2 - \mathrm{i}S_3)/2$, yields

$$\boxed{\begin{aligned} m_k = {} & \tfrac{1}{2}(P_k + Q_k) S_0 + \tfrac{1}{2}(P_k - Q_k) S_1 \\ & + \Re(C_k) S_2 + \Im(C_k) S_3. \end{aligned}} \tag{B5}$$

The four-state instrument is therefore $\boldsymbol{m} = \boldsymbol{A}\boldsymbol{S} + \boldsymbol{\epsilon}$, where row k of $\boldsymbol{A}$ is precisely the four coefficients in Eq. (B5). We stress the scope of this derivation. It is an exact identity for the declared linear electromagnetic response and spatially integrated quadratic observable, not an assertion that the four-state surface creates an ideal waveplate or that ϵ has a known laboratory distribution.

2. Passivity of the modeled analyzer and physical feasibility

At each switch state the two complex basis fields define a 2×2 Hermitian Gram matrix

$$\mathbf{G}_k = \begin{pmatrix} P_k & C_k \\ C_k^* & Q_k \end{pmatrix} \succeq 0. \tag{B6}$$

Cauchy–Schwarz implies $|C_k|^2 \leq P_k Q_k$. In Stokes coordinates, this enforces

$$A_{k0} \geq \sqrt{A_{k1}^2 + A_{k2}^2 + A_{k3}^2}, \tag{B7}$$

provided the indices $0, 1, 2, 3$ refer to the columns defined above. Thus every physical row lies in the Lorentz-like positive cone of nonnegative polarization analyzer responses; a numerical candidate violating Eq. (B7) indicates an extraction or convention fault, not exceptionally good polarization sensitivity. The inequality is a standard consequence of positive semidefinite coherency/Gram matrices[6]; the application-specific aspect here is its evaluation directly from the local HFSS field integrals.

For channel-to-channel comparison, define $\widetilde{A}_{kj} = A_{kj}/A_{k0}$ whenever $A_{k0} > 0$. This operation factors out each state's total gain and reveals its normalized polarization analyzer vector $\boldsymbol{p}_k = (\widetilde{A}_{k1}, \widetilde{A}_{k2}, \widetilde{A}_{k3})$. The bound above becomes $\|\boldsymbol{p}_k\|_2 \leq 1$. In particular, $\eta_3 = \max_k |\widetilde{A}_{k3}|$ is a convenient circular-Stokes *sensitivity coefficient*, but it is not circular dichroism, handedness extinction, or Stokes reconstruction precision. Maximizing η_3 alone can worsen the independence of the four analyzer vectors.

3. Observability, conditioning and uncertainty transfer

The ideal four-measurement inverse exists when $\mathrm{rank}(\boldsymbol{A}) = 4$. For a full-rank square operator, the formal reconstruction is $\widehat{\boldsymbol{S}} = \boldsymbol{A}^{-1}\boldsymbol{m}$. A numerical implementation normally uses a singular-value decomposition, optionally with weighted least squares, to avoid uncontrolled inversion of nearly singular matrices. In the presence of additive measurement and calibration perturbations,

$$\widehat{\boldsymbol{S}} - \boldsymbol{S} = \widehat{\boldsymbol{A}}^{-1}(\boldsymbol{\epsilon} - \Delta\boldsymbol{A}\boldsymbol{S}), \qquad \widehat{\boldsymbol{A}} = \boldsymbol{A} + \Delta\boldsymbol{A}. \tag{B8}$$

It follows that, for a nonsingular $\widehat{\boldsymbol{A}}$,

$$\|\delta\boldsymbol{S}\|_2 \leq \left\| \widehat{\boldsymbol{A}}^{-1} \right\|_2 \left(\|\boldsymbol{\epsilon}\|_2 + \|\Delta\boldsymbol{A}\|_2 \, \|\boldsymbol{S}\|_2 \right). \tag{B9}$$

This bound clarifies why reported matrix conditioning and mesh-transfer error are different quantities: a well-conditioned inverse can still produce inaccurate estimates when the calibration operator changes between simulations or measurements. With a known measurement-noise covariance $\boldsymbol{\Sigma}_m$, the first-order covariance of the reconstructed Stokes vector is $\boldsymbol{\Sigma}_S \simeq \boldsymbol{A}^{-1}\boldsymbol{\Sigma}_m\boldsymbol{A}^{-\mathsf{T}}$ when the matrix itself is taken as known. Calibration uncertainty requires additional terms and cannot be inferred solely from a stipulated percentage of intensity noise.

The geometric viewpoint is equally useful. Four normalized analyzer vectors define a tetrahedron inside the Poincaré unit ball. If their points approach a plane, the smallest singular value collapses and at least one Stokes direction becomes poorly observable; tetrahedral design and conditioning are well-established polarimetry concepts[5,7]. We use these principles as design diagnostics, not as a newly invented condition for measuring Stokes vectors. Detailed equivalences between Gram positivity, analyzer geometry, and perturbation bounds are provided in the Supplementary Material.

4. Derived design tests: admissibility, information bandwidth, and calibration transfer

The field-overlap construction supplies three directly checkable results for candidate screening. First, Cauchy–Schwarz forces each gain-normalized analyser row to remain inside the unit Poincaré ball [Eq. (B7)]; this is an extraction-consistency test, not a new universal physical law. Second, the frequency-domain *nominal information set* for thresholds $\kappa_{\max}$ and $\eta_{\min}$ is

$$\begin{aligned}\mathcal{B}_{\mathrm{nom}} = \{f :\ &\operatorname{rank} A(f) = 4,\\ &\kappa_2(\widetilde{A}(f)) \le \kappa_{\max},\ \eta_3(f) \ge \eta_{\min}\}.\end{aligned} \tag{B10}$$

A connected component of $\mathcal{B}_{\mathrm{nom}}$ is only a *candidate* information bandwidth; measured sensitivity, finite-board transmission and independent-mesh agreement are additional necessary tests. Third, Eq. (B9) provides an exact algebraic route from calibration drift to an error bound. These relationships were checked against archived finite complex-field data and synthetic Jones vectors. Their novelty is their particular combination in a receiver-aware HFSS screening workflow, not the underlying algebra of Stokes vectors, matrix norms or positive semidefinite quadratic forms.

5. Derived equator restriction error and ideal analyser conditioning

Two analytical baselines are useful in interpreting the early partial-Stokes estimates. Let a fully polarized state have azimuth ψ and ellipticity angle $\chi \in [-\pi/4, \pi/4]$. Its normalized Stokes vector can be written $\boldsymbol{s} = (\cos 2\chi \cos 2\psi, \cos 2\chi \sin 2\psi, \sin 2\chi)$. A procedure that forces $S_3 = 0$ and projects the measured vector onto the nearest equatorial direction incurs the geodesic angular separation

$$\epsilon_{\mathrm{lin}} = \cos^{-1}(\cos 2\chi) = 2|\chi|, \quad |\chi| \le \pi/4, \tag{B11}$$

where the angle is a *Poincaré-sphere angular error*, not a physical antenna pointing angle. The relation follows directly from the inner product of the normalized state and its equatorial projection, and assumes full polarization. It quantifies the consequence of ignoring an unmeasured circular coordinate even when the linear components have been determined exactly. The 0.887 linear-Stokes magnitude reported in the historical estimate does not establish whether the remainder is circularly polarized or partially polarized: therefore Eq. (B11) must not be used to assign a unique historical χ.

A second analytic benchmark is the tetrahedral four-analysis instrument. Take four unit analyzer vectors $\boldsymbol{n}_k$ that satisfy $\boldsymbol{n}_i \cdot \boldsymbol{n}_j = -1/3$ for $i \ne j$ and define ideal equally weighted rows $a_k = \frac{1}{2}(1, \boldsymbol{n}_k^{\mathsf{T}})$. Then $A^{\mathsf{T}}A = \mathrm{diag}(1, 1/3, 1/3, 1/3)$ and

$$\kappa_2(A_{\mathrm{tetra}}) = \sqrt{3}. \tag{B12}$$

The equality is established polarimeter theory and here serves solely as a transparent mathematical comparator. Suppressing the circular coordinates of the analyzer vectors collapses the smallest singular value and eventually the matrix rank. Thus a nominal rank-four matrix with a very small singular value is not close to a tetrahedral instrument just because all four Stokes coefficients are nonzero. The numerical benchmarks are reproducible from these expressions and the archived analysis script.

Appendix C: Related software and industry engagement

The desktop *Rydberg Workbench* prototype and deployable mechanical design illustrate work toward a future integrated sensor. Their development was documented separately and they are not relied upon as validation of metasurface polarimetry. Industrial engagement with DSTG and Panop informed application awareness; no unverified endorsement or technical result is attributed to those organisations.

AUTHOR DECLARATIONS

Conflict of interest. Author declaration to be completed before submission.

Author contributions. R. Verghese: research design, electromagnetic simulations, analysis and manuscript drafting; collaborators' roles subject to confirmation.

Data availability. The accompanying reproducibility archive includes native HFSS-derived Jones and Stokes numerical tables, the C000/C033/C015 optimisation parameters, multi-frequency dual-mesh results, finite-field

verification summary and plot-generating scripts. Native AEDT project directories and original laboratory files should be preserved in the corresponding project archive; this paper package alone does not duplicate all historical full-size HFSS databases.

Research scope. Simulations are electromagnetic-only and contain no synthetic detector noise. No completed Rydberg optical readout response or experimental full-Stokes data are reported.

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